Assignment 1

(1)

(1) Determine whether the following composition laws are associative, whether they are commutative and whether they have an identity
   (a) $(x, y) \mapsto x - y$ on $\mathbb{Z}$;
   (b) $(x, y) \mapsto x + y + xy$ on $\mathbb{Q} \setminus \{-1\}$.
Does any of them define a group structure? justify!

(a)

Associativity : No !

$((x,y),z) = (x-y,z) = (x-y)-z = x-y-z$

$(x,(y,z)) = (x, y-z) = x-(y-z) = x-y+z$

$x-y-z \neq x-y+z$ in $\mathbb{Z}$

Commutative : No !

$(x,y) = x-y$

$(y,x) = y-x$

$x-y \neq y-x$ in $\mathbb{Z}$

Identity : No !

suppose we have the identity called e

then it satisfies : $\forall x \in \mathbb{Z}$ $\begin{cases} (x,e) = x & -① \\ (e,x) = x & -② \end{cases}$

① $\Leftrightarrow$ $x-e = x \Rightarrow e = 0$

② $\Leftrightarrow$ $e-x = x \overset{①}{\Rightarrow} x = 0$

It is contradition to "$\forall x \in \mathbb{Z}$"

So $\nexists$ identity.

(b)

Associativity: Yes! both on $\mathbb{Z}$ and $\mathbb{Q} \setminus \{-1\}$

$$((x,y),z) = (x+y+xy, z) = (x+y+xy) + z + (x+y+xy)z$$

$$= x+y+z+xy+yz+zx+xyz$$

$$(x,(y,z)) = (x,(y+z+yz)) = x+(y+z+yz)+x(y+z+yz)$$

$$= x+y+z+xy+yz+zx+xyz$$

So $((x,y),z) = (x,(y,z))$

Commutative: Yes! both on $\mathbb{Z}$ and $\mathbb{Q} \setminus \{-1\}$

$$(x,y) = x+y+xy$$

$$(y,x) = y+x+yx = x+y+yx$$

$\downarrow$

$x,y,z \in \mathbb{Z}$ or $\mathbb{Q} \setminus \{-1\}$

So $(x,y) = (y,x)$

Identity : the identity is 0

since : $\forall x \in \mathbb{Z}$ or $\mathbb{Q}\setminus\{-1\}$: $\begin{cases} (x,0) = x+0+x\cdot 0 = x \\ (0,x) = 0+x+0\cdot x = x \end{cases}$

$\Rightarrow (x,0) = x = (0,x)$

So 0 is the identity

Only (b) on $\mathbb{Q}\setminus\{-1\}$ determine a group structure.

(a) : can't because not associative

(b) : $(x,y) \longmapsto x+y+xy$ is $G \times G \longrightarrow G$

We only need inverse :

$(x, x^{-1}) = 0 \Rightarrow x + x^{-1} + xx^{-1} = 0$

$\rightarrow x^{-1} = \frac{-x}{1+x}$

Inverse exist only for $x \neq -1$

Therefore can't on $\mathbb{Z}$, only can on $\mathbb{Q}\setminus\{-1\}$ ∎

(2) Prove that a group G that has the property that $a^2 = e$ for all $a \in G$ is abelian.

(2)

proof:

since $\forall a \in G : a^2 = a \cdot a = e$

then $a^{-1}(a \cdot a) = a^{-1} \cdot e$

$\Rightarrow (a^{-1} \cdot a)\, a = a^{-1}$

$\Rightarrow a = a^{-1}$

take $x, y \in G$

$xy = x^{-1} y^{-1} = (yx)^{-1} = yx$

so G is abelian ∎

(3) Let a and b be elements of a group G. Prove that the order of ab is equal to the order of ba.

(3)

proof :

Suppose the order of ab is n

that is : n is the minimal number s.t. $(ab)^n = e$

now to prove the order of ba is also n :

- To show $(ba)^n = e$

$$(ba)^n = \underbrace{(ba)(ba)\cdots(ba)}_{n} = b\underbrace{(ab)(ab)\cdots(ab)}_{n-1}a$$

$$= b\underbrace{(ab)(ab)\cdots(ab)}_{n-1}a(bb^{-1})$$

$$= b\underbrace{(ab)(ab)\cdots(ab)(ab)}_{n}b^{-1}$$

$$= b(ab)^n b^{-1}$$

$$= b \cdot e \cdot b^{-1}$$

$$= e$$

- To show n is the smallest :

suppose $\exists\, m < n \;/\; (ba)^m = e$

Since $(ba)^m = \underbrace{(ba)(ba)\cdots(ba)}_{m}$

$= b\underbrace{(ab)(ab)\cdots(ab)}_{m-1}a$

$= b\underbrace{(ab)(ab)\cdots(ab)}_{m-1}a(bb^{-1})$

$= b\underbrace{(ab)(ab)\cdots(ab)(ab)}_{m}b^{-1}$

$= b(ab)^m b^{-1}$

then $b(ab)^m b^{-1} = e$

$\Rightarrow b(ab)^m b^{-1} \cdot b = e \cdot b$

$\Rightarrow b(ab)^m = b$

$\Rightarrow b^{-1}b(ab)^m = b^{-1} \cdot b$

$\Rightarrow (ab)^m = e$

it is contradiction to order$(ab) = n$ $(m < n)$

Therefore: order$(ba) = n$ ∎

(4) Let G be a group. Prove that the following subsets of G are subgroups.
(a) $H_1 \cap H_2$ where $H_i \leq G$.
(b) The *centraliser* of $a \in G$ defined by $C_G(a) = \{x \in G \mid ax = xa\}$.
(c) The *centre* of G defined by $Z(G) = \{x \in G \mid xy = yx, \forall y \in G\}$.

(4)

(a)

proof : we have the property that :

property : $H \leq G$ iff

① $e \in H$

② $\forall a, b \in H \Rightarrow ab \in H$

③ $\forall a \in H \Rightarrow a^{-1} \in H$

so we need to show ①②③ with $H_1 \cap H_2$

① : since $H_1 \leq G$, $H_2 \leq G$

then $e \in H_1$, $e \in H_2$

$\Rightarrow e \in H_1 \cap H_2$

② take $a, b \in H_1 \cap H_2$

in particular $\begin{cases} a, b \in H_1 \overset{H_1 \leq G}{\Rightarrow} ab \in H_1 \\ a, b \in H_2 \overset{H_2 \leq G}{\Rightarrow} ab \in H_2 \end{cases}$

then $ab \in H_1 \cap H_2$

③ take $x \in H_1 \cap H_2$

in particular $x \in H_1 \wedge x \in H_2$

$\Rightarrow x^{-1} \in H_1 , x^{-1} \in H_2$

$H_i \leq G$

$\Rightarrow x^{-1} \in H_1 \cap H_2$

Therefore $H_1 \cap H_2 \leq G$ ∎

(b)

proof: the same, we need to show ①②③ with $C_G(a)$

①: since $e \cdot a = a = a \cdot e$

then $e \in C_G(a)$

②: take $x, y \in C_G(a)$

then $\begin{cases} xa = ax & -① \\ ya = ay & -② \end{cases}$

then $(xy)a = x(ya) \overset{①}{=} x(ay)$

$= (xa)y$

$\overset{②}{=} (ax)y$

$= a(xy)$

so $(xy) \in C_G(a)$

③ take $x \in C_G(a)$

then $xa = ax \quad -①$

① $\Rightarrow a = x^{-1}ax$

then $ax^{-1} = (x^{-1}ax)x^{-1}$

$= (x^{-1}a)(xx^{-1})$

$= x^{-1}a$

so $x^{-1} \in C_G(a)$

Therefore: $C_G(a) \leq G$ ∎

(G)

① since $e \cdot y = y = y \cdot e \quad \forall y \in G$

then $e \in Z(G)$

② take $a, b \in Z(G)$

then $\begin{cases} ax = xa & \forall x \in G \quad -① \\ by = yb & \forall y \in G \quad -② \end{cases}$

then $(ab)z \overset{②}{=} a(bz) = a(zb) = (az)b \overset{①}{=} (za)b$

$= z(ab)$

then $ab \in Z(G)$

③ take $a \in Z(G)$

then $ax = xa \quad \forall x \in G$

then $a^{-1}x = a^{-1}(axa^{-1}) = (a^{-1}a)(xa^{-1})$

$= xa^{-1} \quad \forall x \in G$

so $a^{-1} \in Z(G)$

Therefore $Z(G) \leq G$

(5) Prove that the inverse of a permutation matrix is its transpose.

(5)

Def: permutation matrix is a $n \times n$ matrix that every row, column has only one "1"

proof: Suppose we have a permutation matrix P.

to show $P^{-1} = P^T$ we just need to show that $PP^T = I = P^TP$

① $PP^T = I$:

Suppose $P \in M_{n\times n}$ $\quad P = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{i1} & P_{i2} & \cdots & P_{in} \\ P_{n1} & \cdots & & P_{nn} \end{bmatrix} \leftarrow i$ $\quad P^T = \begin{bmatrix} P_{11} & P_{21} & \cdots & P_{n1} \\ P_{12} & P_{22} & & \\ \vdots & & & \\ P_{1n} & \cdots & & P_{nn} \end{bmatrix}$ ($\vec{j}$: $P_{j1}, P_{j2}, \dots, P_{jn}$)

then $(PP^T)_{ij} = P_{i1}P_{j1} + P_{i2}P_{j2} + \cdots + P_{in}P_{jn}$

If $i \neq j$:

then $(PP^T)_{ij} = \underbrace{P_{i1}P_{j1}}_{=0} + \underbrace{P_{i2}P_{j2}}_{=0} + \cdots + \underbrace{P_{in}P_{jn}}_{=0} = 0$

If $i = j$:

then $(PP^T)_{ij} = P_{i1}P_{j1} + P_{i2}P_{j2} + \cdots + P_{in}P_{jn}$

$= P_{i1}^2 + P_{i2}^2 + \cdots + P_{in}^2$ (one of them = 1, other = 0)

$= 1$

So $PP^T = I$

Similary : $P^TP = I$

Therefore : $P^{-1} = P^T$ ∎